SSL Certificate Deployment

Plan of Action (POA)

ASNB Production Environment

Date: 06 July 2026

Domain: asnbcc.tmone.com.my

This POA covers certificate generation and deployment for ASNB using Method A (Python automation) and Method B (Bash script). Includes pre-checks, deployment steps, verification, rollback, and sign-off.

1. Document Control

Domain: asnbcc.tmone.com.my
Keystore Alias: ameyo-server
Keystore Password: (configured in configs/asnb.yaml)
Servers: asnb-app-db (AppServer, Reports, CRM), asnb-cs (Asterisk)

2. Pre-Requisites

- [ ] GlobalSign EML certificate email received
- [ ] Private key file available (must match certificate)
- [ ] SSH access: dacx@asnb-app-db, root@asnb-cs
- [ ] Maintenance window approved
- [ ] Backup of existing certificates taken
- [ ] configs/asnb.yaml and deploy.conf reviewed

3. Method A - Python Automation (Recommended)

Location: ssl-cert-tool folder on Windows laptop

Step A1: Setup (one-time)

```
cd ssl-cert-tool
setup.bat
```

Step A2: Inspect customer cert

```
python ssl-auto.py inspect --input "ASNB CERT" --key "ASNB CERT/asnbcc.tmone.com.my.key"
```

Expected: Key/Cert match PASS, CN matches domain, expiry ~2027-01-16

Step A3: Generate deployment files

```
python ssl-auto.py generate --input "ASNB CERT" --key "asnbcc.tmone.com.my.key" --alias "ameyo-server" --password "<
```

Step A4: Validate and plan

```
python ssl-auto.py validate --config asnb
python ssl-auto.py plan --config asnb --certs output/asnbcc.tmone.com.my
```

Step A5: Dry-run deploy

```
python ssl-auto.py deploy --config asnb --certs output/asnbcc.tmone.com.my --dry-run
```

Step A6: Production deploy

```
python ssl-auto.py deploy --config asnb --certs output/asnbcc.tmone.com.my
```

Deploy order: Asterisk (asnb-cs) -> CRM -> AppServer/Reports (asnb-app-db)

Step A7: Full automation (one command)

```
python ssl-auto.py run --config asnb --input "ASNB CERT" --key "asnbcc.tmone.com.my.key" --alias "ameyo-server"
```

4. Method B - Bash Script (No Python on Server)

Generate certs on Windows (Method A Steps A1-A3), then deploy from Linux jump host.

Step B1: Copy certs to jump host

```
scp -r output/asnbcc.tmone.com.my/ user@jump-host:/tmp/certs/
```

Step B2: Dry-run

```
./deploy.sh --certs /tmp/certs/asnbcc.tmone.com.my --config deploy.conf --dry-run
```

Step B3: Deploy

```
./deploy.sh --certs /tmp/certs/asnbcc.tmone.com.my --config deploy.conf
```

Step B4: Per-server deploy (optional)

```
./deploy.sh --certs /tmp/certs/asnbcc.tmone.com.my --config deploy.conf --server ASNB_CS
./deploy.sh --certs /tmp/certs/asnbcc.tmone.com.my --config deploy.conf --server ASNB_APP_DB
```

5. Deployment Paths

File	Remote Path	Service
asnbcc.jks	/dacx/ssl2025/asnbcc.jks	AppServer, Reports
crm.crt	/dacx/ssl2025/crm.crt	CRM
asnbcc.tmone.com.my.key	/dacx/ssl2025/	CRM
chain.crt	/dacx/ssl2025/chain.crt	CRM
asterisk.crt/key/pem	/dacx/var/ameyo/.../asterisk/keys/	Asterisk

6. Post-Deploy Verification

```
keytool -list -keystore /dacx/ssl2025/asnbcc.jks -storepass <password> -alias ameyo-server
openssl x509 -in /dacx/ssl2025/crm.crt -noout -subject -dates
openssl x509 -in .../asterisk/keys/asterisk.crt -noout -subject -dates
```

- [ ] JKS opens with configured password
- [ ] New expiry date confirmed
- [ ] AppServer HTTPS OK
- [ ] Reports HTTPS OK
- [ ] CRM HTTPS OK
- [ ] Asterisk TLS OK
- [ ] Stakeholder sign-off

7. Rollback

```
python ssl-auto.py deploy --config asnb --rollback --server asnb_app_db
./deploy.sh --config deploy.conf --rollback --server ASNB_APP_DB
```

Tool creates automatic backups before overwriting files.

8. Important Notes

- Tool copies certificate files only; does NOT edit tomcat.conf or httpd.conf
- Always dry-run before production deploy
- AppServer and Reports share same JKS file
- GUI option: double-click ssl-gui.bat
- Credentials stored in configs/asnb.yaml - do not share externally

9. Sign-Off

Role	Name	Signature / Date
Performed By		
Reviewed By		
Approved By		
Date Completed		